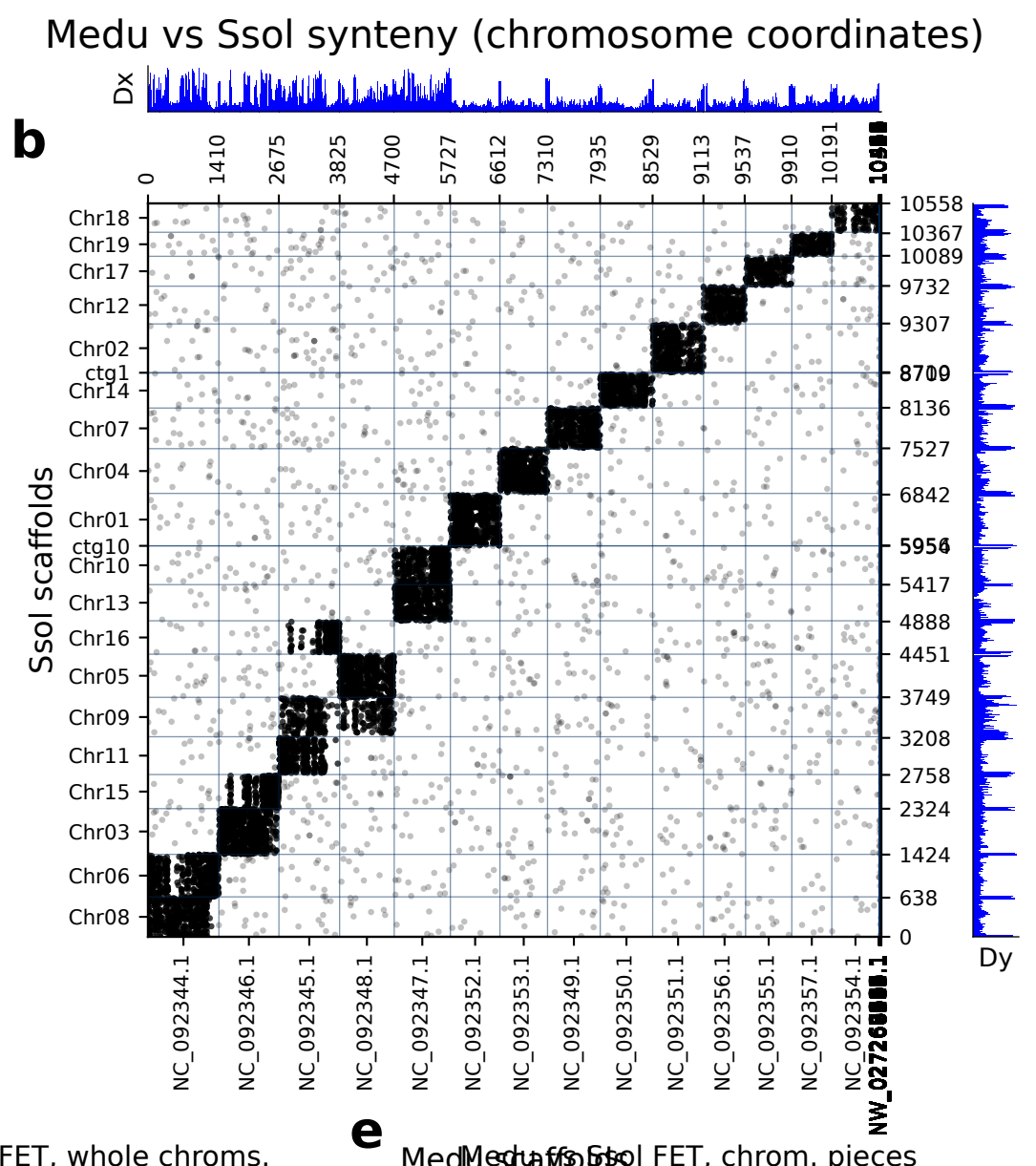
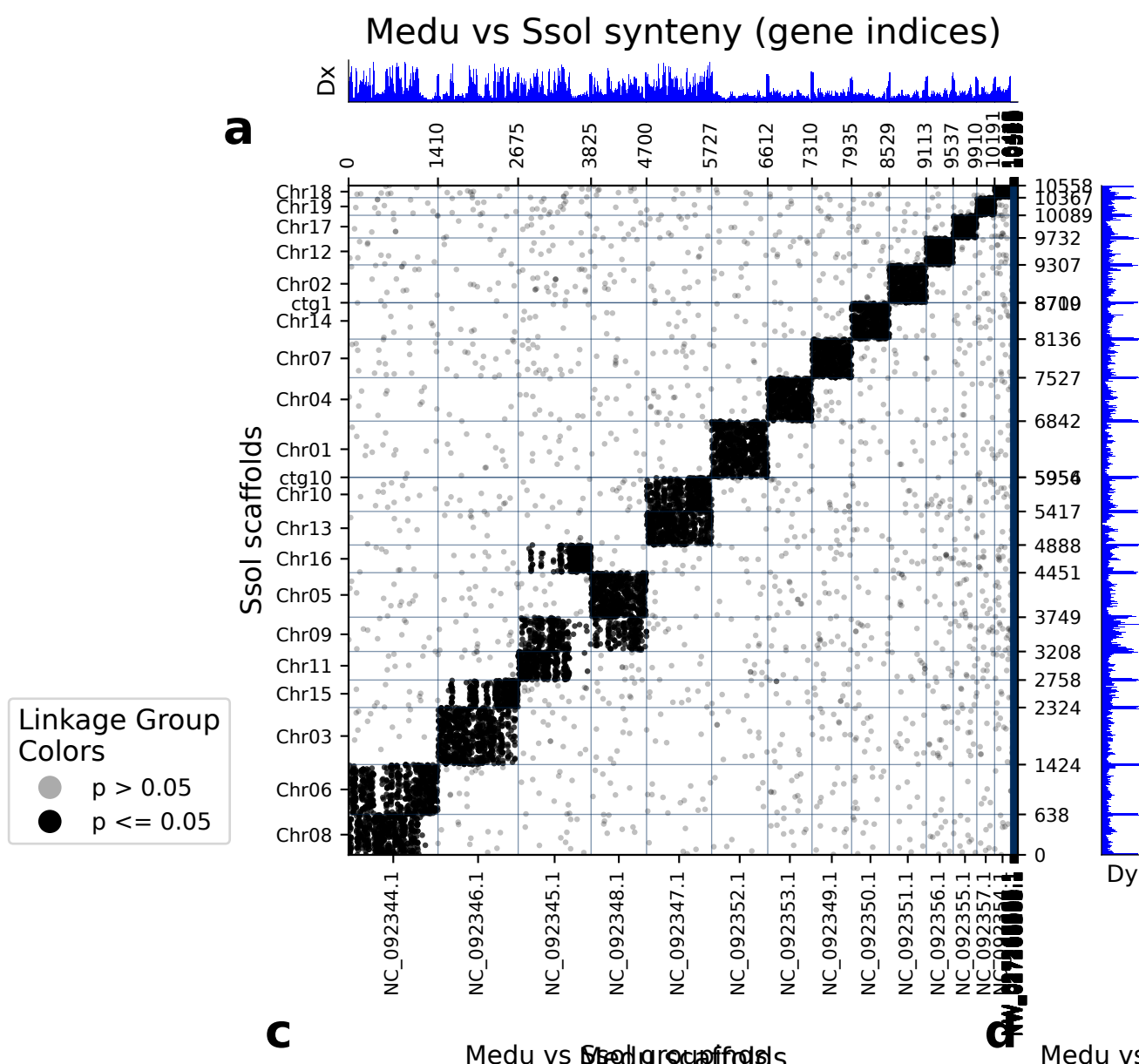


diamond: 10559 orthologs on 121 Medu scaffolds and 21 Ssol scaffolds.



C Medu vs Medu scaffolds

G Medu vs Ssol FET, whole chroms.

	Medu scaf	Ssol scaf	p (FET)	Count
0	NC_092352.1	Chr01	0.00e+00	782
1	NC_092346.1	Chr03	0.00e+00	764
2	NC_092344.1	Chr06	0.00e+00	692
3	NC_092353.1	Chr04	0.00e+00	582
4	NC_092348.1	Chr05	0.00e+00	577
5	NC_092344.1	Chr08	0.00e+00	539
6	NC_092349.1	Chr07	0.00e+00	493
7	NC_092351.1	Chr02	0.00e+00	485
8	NC_092350.1	Chr14	0.00e+00	483
9	NC_092347.1	Chr13	0.00e+00	466
10	NC_092347.1	Chr10	0.00e+00	417
11	NC_092345.1	Chr11	7.40e-284	365
12	NC_092345.1	Chr16	4.62e-259	345
13	NC_092346.1	Chr15	2.31e-238	341
14	NC_092356.1	Chr12	0.00e+00	332
15	NC_092355.1	Chr17	0.00e+00	271
16	NC_092345.1	Chr09	1.29e-97	247
17	NC_092357.1	Chr19	2.79e-274	196
18	NC_092348.1	Chr09	7.37e-59	175
19	NC_092354.1	Chr18	6.46e-176	130

e Medufo FET, chrom. pieces

	Medu scaf	Ssol scaf	p (FET)	Count
0	NC_092352.1 : 0-end	Chr01 : 0-end	0.00e+00	782
1	NC_092346.1 : 0-end	Chr03 : 0-end	0.00e+00	764
2	NC_092344.1 : 0-end	Chr06 : 0-end	0.00e+00	692
3	NC_092353.1 : 0-end	Chr04 : 0-end	0.00e+00	582
4	NC_092348.1 : 0-end	Chr05 : 0-end	0.00e+00	577
5	NC_092344.1 : 0-end	Chr08 : 0-end	0.00e+00	539
6	NC_092349.1 : 0-end	Chr07 : 0-end	0.00e+00	493
7	NC_092351.1 : 0-end	Chr02 : 0-end	0.00e+00	485
8	NC_092350.1 : 0-end	Chr14 : 0-end	0.00e+00	483
9	NC_092347.1 : 0-end	Chr13 : 0-end	0.00e+00	466
10	NC_092347.1 : 0-end	Chr10 : 0-end	0.00e+00	417
11	NC_092345.1 : 0-end	Chr11 : 0-end	7.40e-284	365
12	NC_092345.1 : 0-end	Chr16 : 0-end	4.62e-259	345
13	NC_092346.1 : 0-end	Chr15 : 0-end	2.31e-238	341
14	NC_092356.1 : 0-end	Chr12 : 0-end	0.00e+00	332
15	NC_092355.1 : 0-end	Chr17 : 0-end	0.00e+00	271
16	NC_092345.1 : 0-end	Chr09 : 0-end	1.29e-97	247
17	NC_092357.1 : 0-end	Chr19 : 0-end	2.79e-274	196
18	NC_092348.1 : 0-end	Chr09 : 0-end	7.37e-59	175
19	NC_092354.1 : 0-end	Chr18 : 0-end	6.46e-176	130

(a) depicts the chromosomes/scaffolds of Medu (x-axis) plotted against the chromosomes/scaffolds of Ssol (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10559 orthologs found between 121 Medu scaffolds and 21 Ssol scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.